

Lecture 9

Mitosis



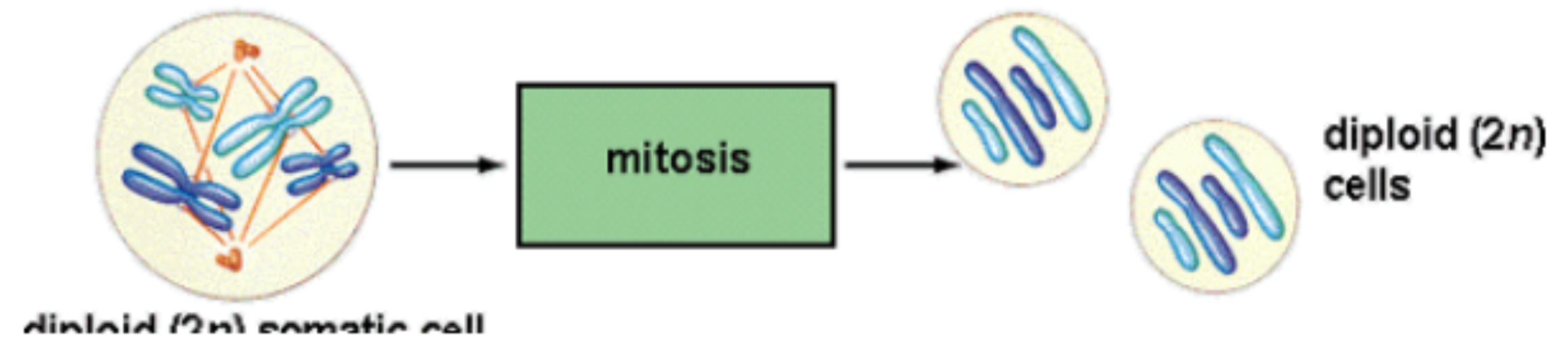
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Eukaryotic cell division

- Mitosis

- Growth
- Repair of damaged tissue
- Genetic stability
- Asexual reproduction



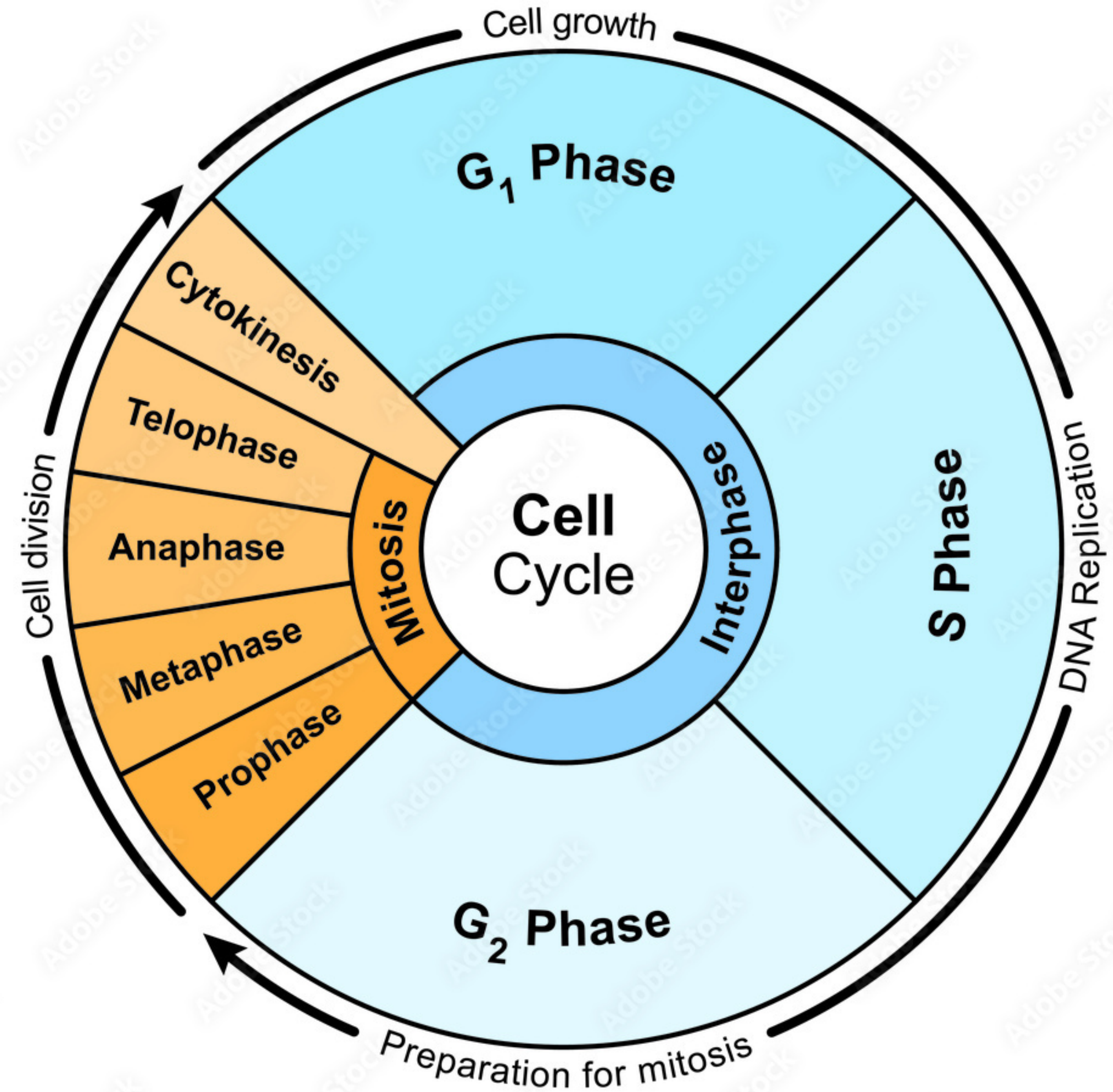
- Meiosis

- Sexual reproduction
- Produces haploid gametes



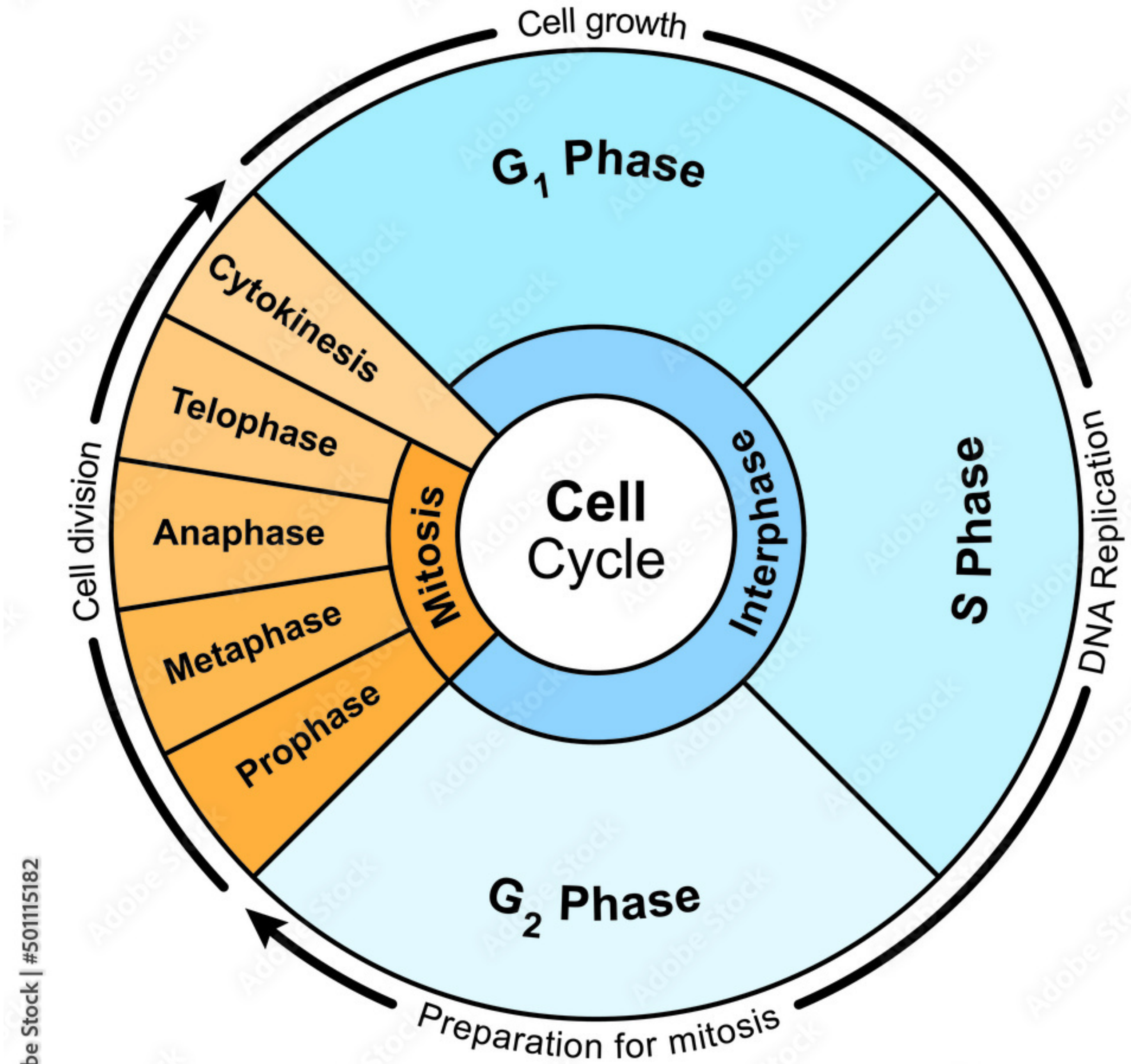
Interphase - G₁

- Synthesis of new proteins and lipids
- Maintenance of cell structure
- Metabolism
- Checkpoint
- Deciding to divide or not



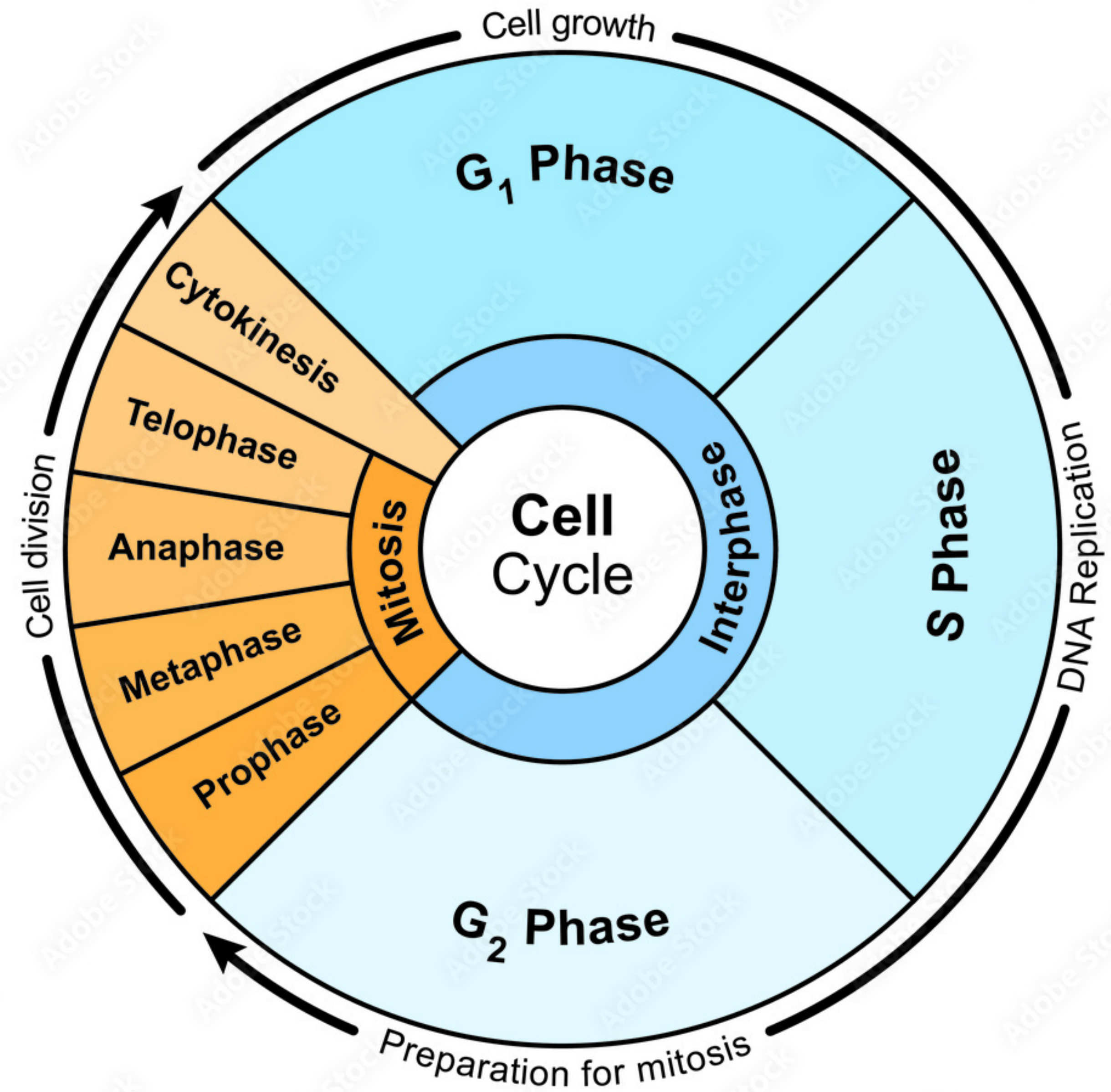
Interphase - S

- DNA replication
- Histone modification
- Cohesion removal



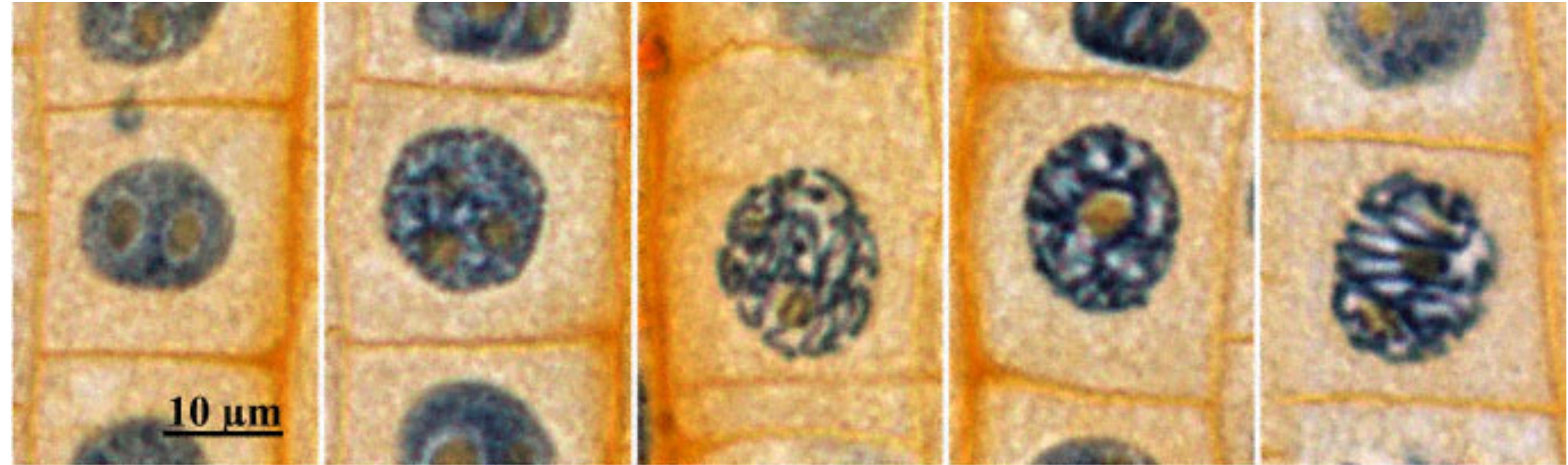
Interphase - G₂

- DNA repair
- Cell growth
- Checkpoint
- Spindle assembly
- Preparation for mitosis



Mitosis - prophase

Interphase Start of prophase Early prophase Middle of prophase Late prophase

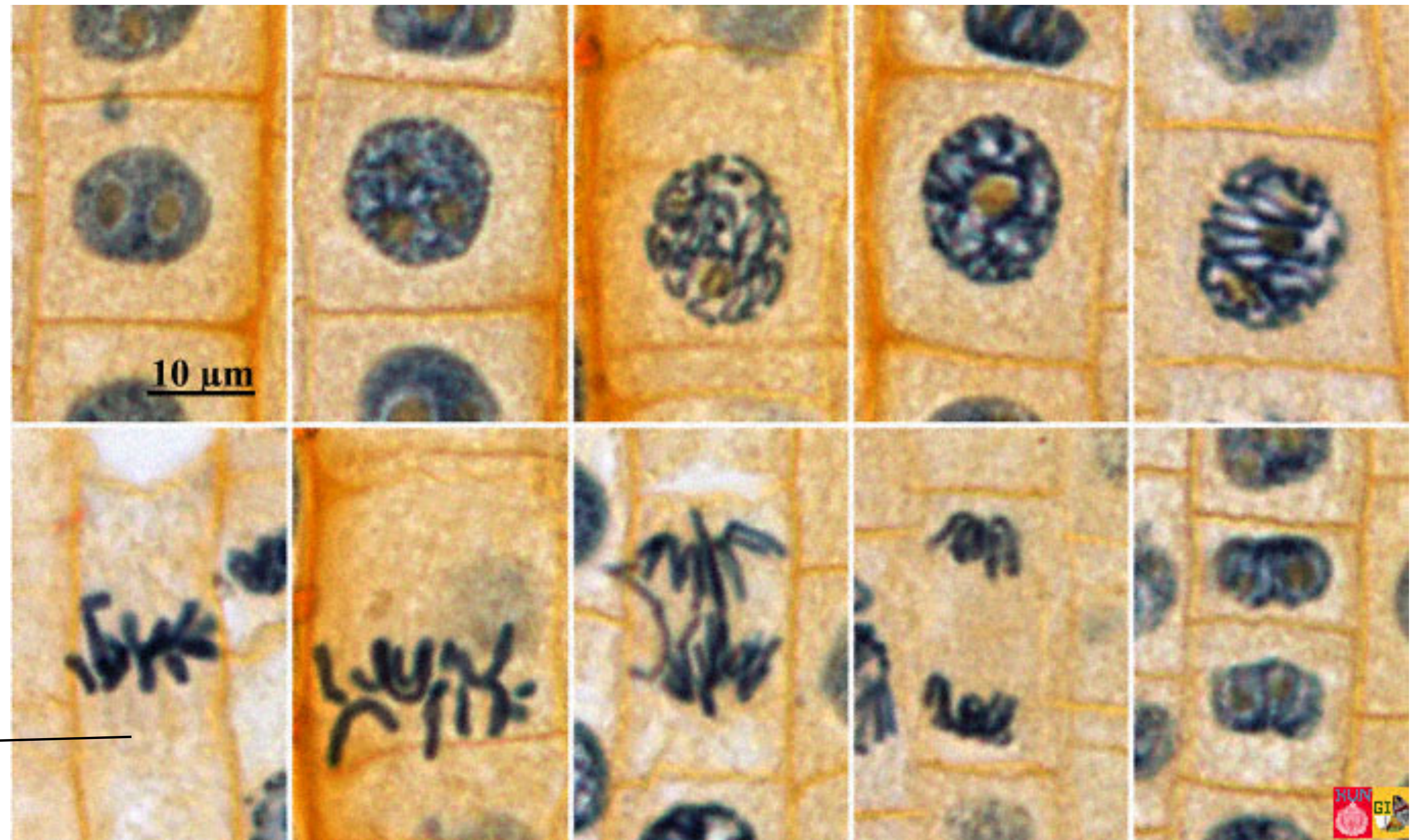


- Chromatin condensation
- Nuclear envelope breakdown
- Spindle formation
- Centrosomes move to opposite poles
- Kinetochore formation

Mitosis - metaphase

- Chromosome alignment
- Spindle attachment
- Tension
- Checkpoint

Metaphase —————



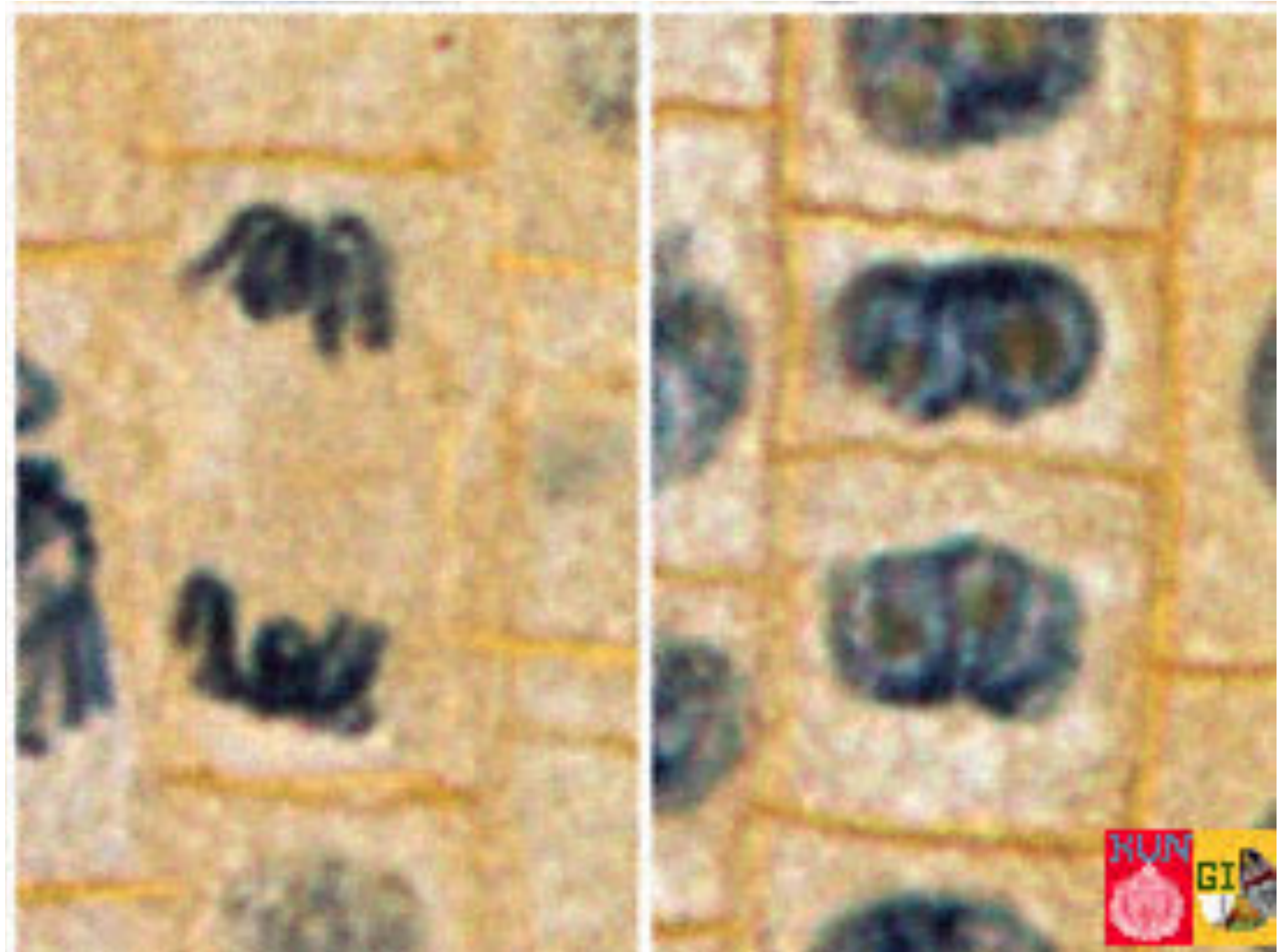
Mitosis - anaphase

- Chromosome separation
- Movement of chromosomes
- Cleavage furrow formation
- Cytokinensis



Mitosis - telophase

- Chromosome decondensation
- Nuclear envelope formation
- Spindle disassembly
- Chromatin remodelling
- Completion of cytokinesis



Lecture 10

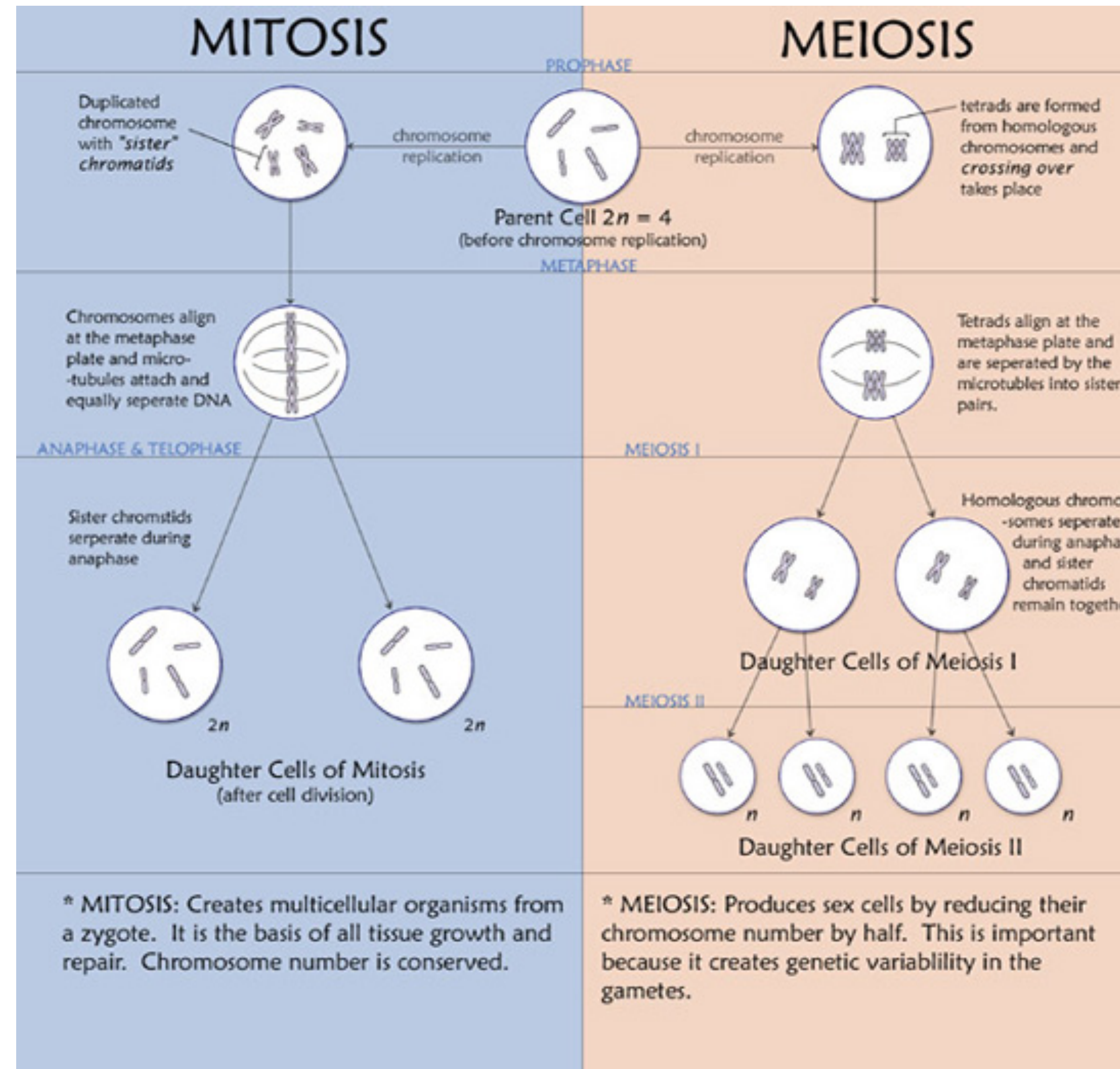
Meiosis



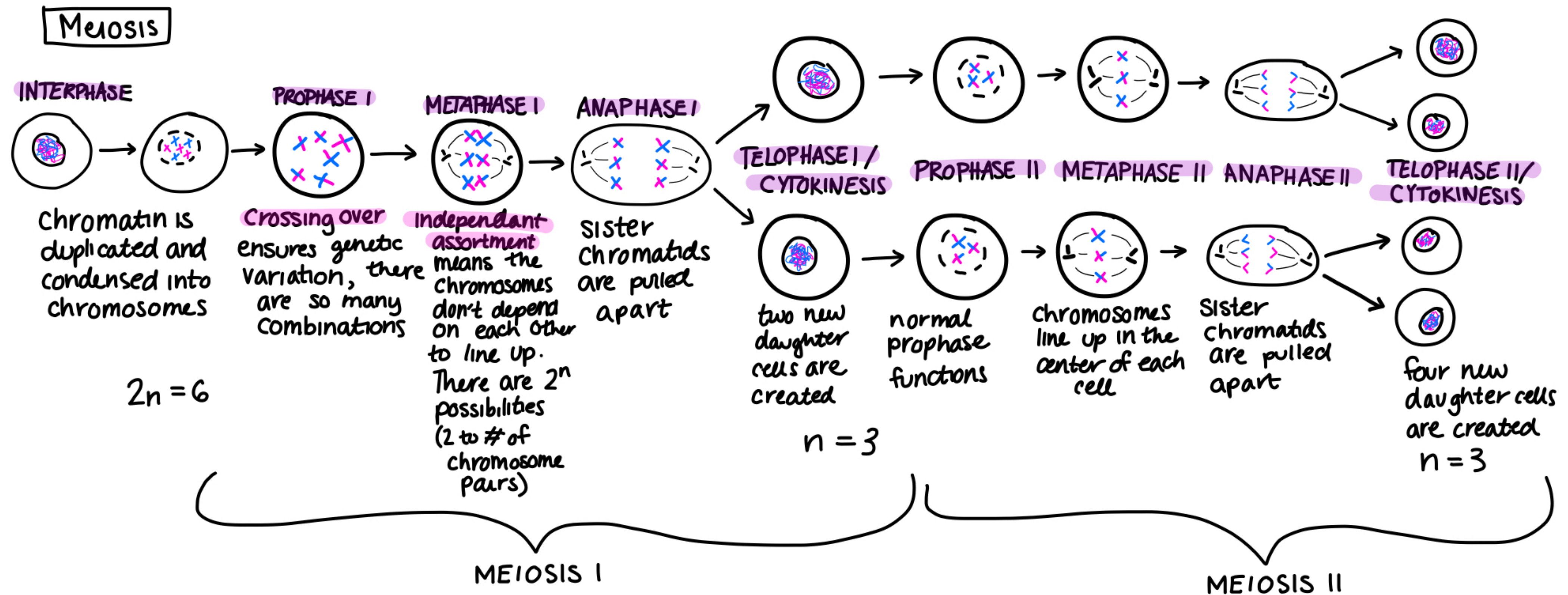
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Comparison of mitosis with meiosis



Overview of the phases of meiosis

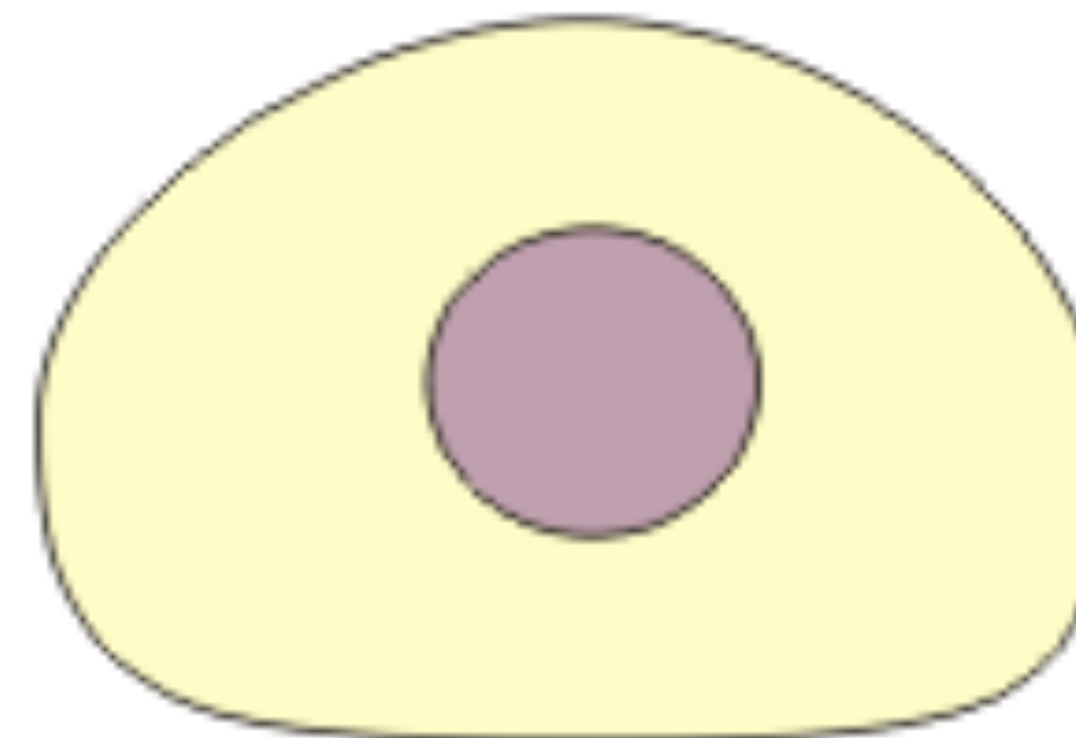


https://commons.wikimedia.org/wiki/File:Process_of_Meiosis.jpg

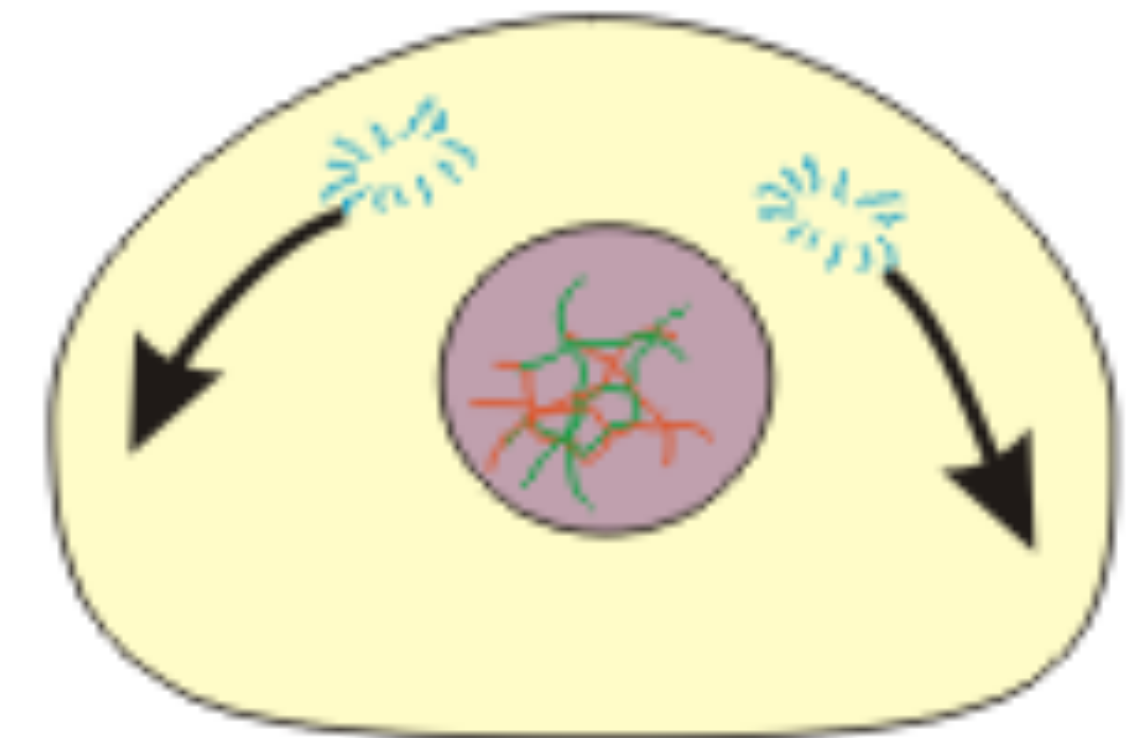


Details of meiosis prophase I

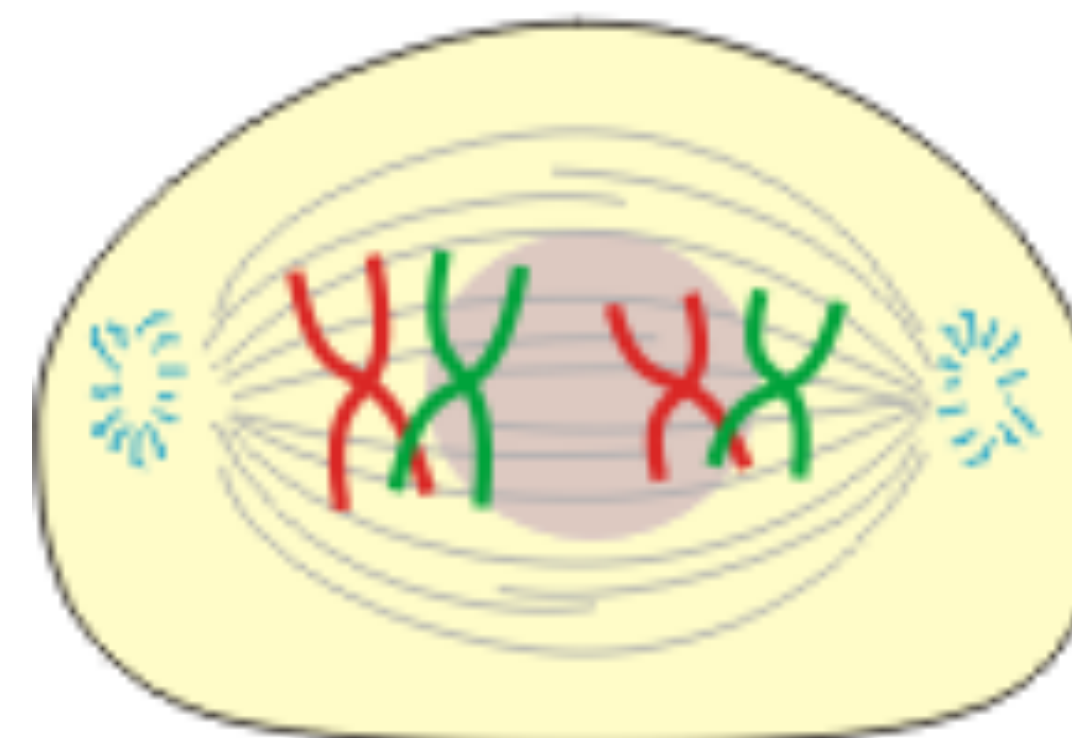
- Chromosome condensation
- Synapsis
- Crossing over
- Chiasmata formation
- Leptotene
- Zygotene
- Pachytene
- Diplotene
- Diakinensis



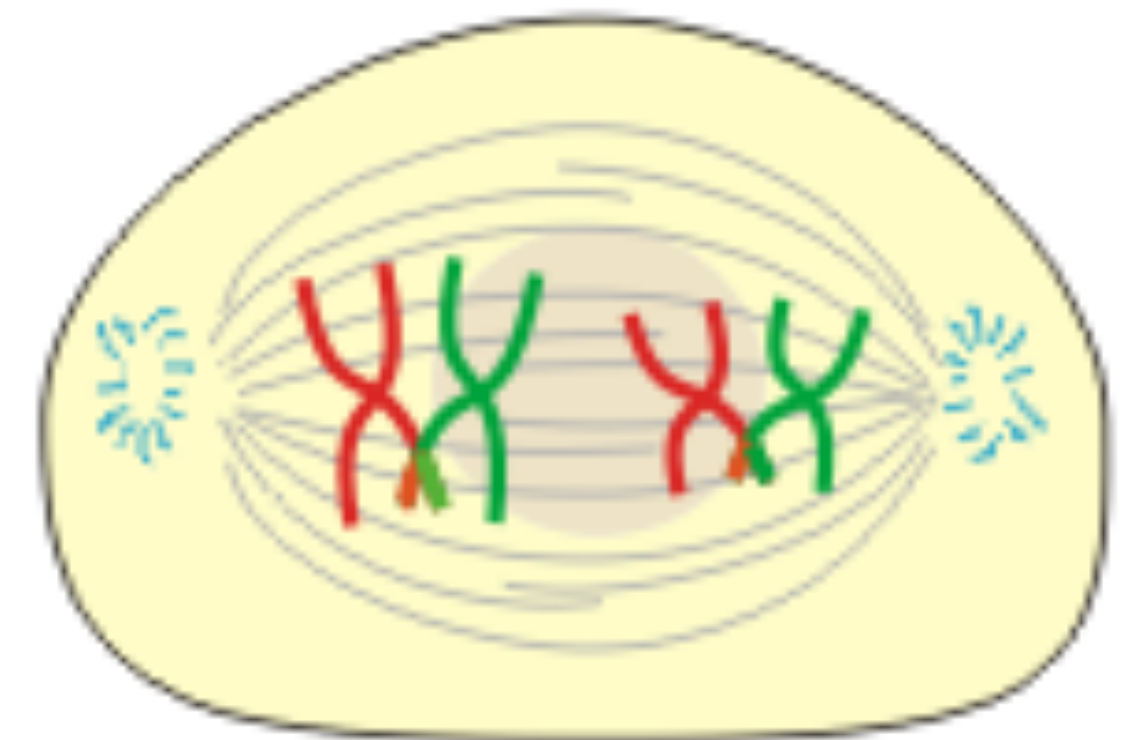
Early prophase I



Middle prophase I



Middle prophase I



Late prophase I